

Lab Assignment 7: Sampling Techniques on Real-World Data

Dataset Requirement:

Use any dataset with at least 1000 records and a mix of categorical and numerical variables (e.g., Titanic dataset, UCI Adult Income dataset, or any Kaggle dataset).

Part A: Probability-Based Sampling Techniques

1. Implement Simple Random Sampling on a real-world dataset (sample size: 10-20%).
2. Implement Systematic Sampling and extract a sample of similar size.
3. Implement Stratified Sampling based on a categorical column (e.g., gender or class).
4. Implement Cluster Sampling by selecting complete groups/clusters.

Part B: Non-Probability-Based Sampling Techniques

5. Implement Convenience Sampling using accessible or first few records.
6. Implement Purposive (Judgmental) Sampling based on specific criteria.
7. Simulate Snowball Sampling starting from a small group and expanding using a linking attribute.
8. Implement Quota Sampling by selecting a fixed number of samples from different categories.